



DIGITALNA FORENZIKA DRUŠTVENIH MEDIJA

FACEBOOK

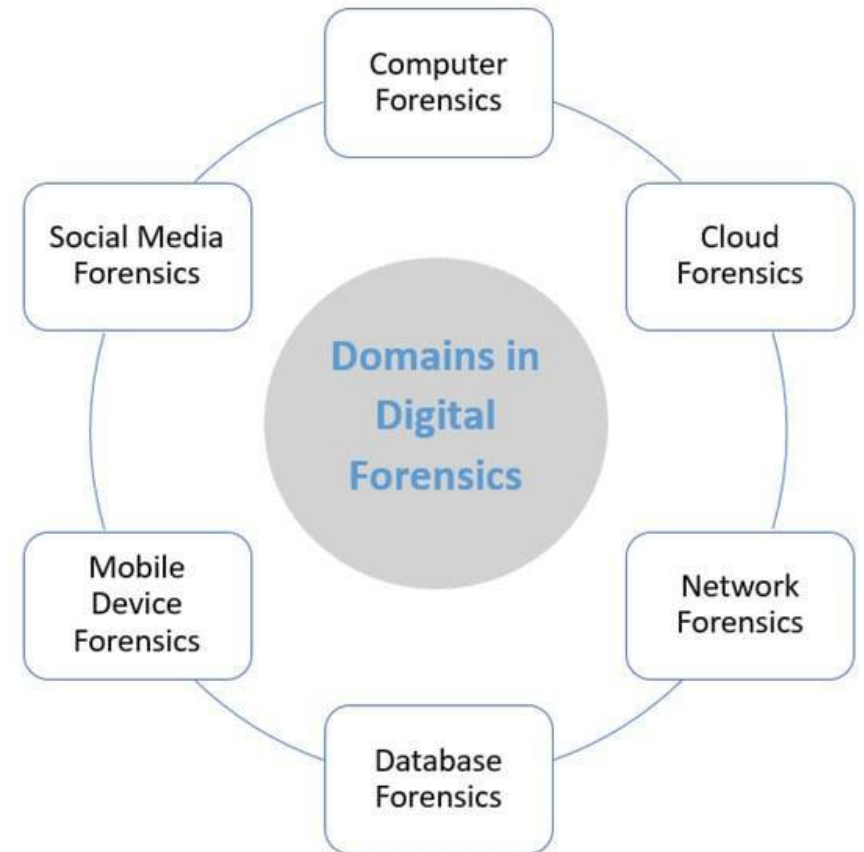


AGENDA

- **Digitalna forenzika društvenih medija**
 - Razumevanje, domeni i ciljevi
 - Faze forenzike društvenih mreža
 - Alati za istraživanje društvenih medija
 - Uticaj veštačke inteligencije
- **Facebook**
 - Definicija, pravila i mogućnosti
 - Forenzika Facebook naloga, AI alati
- **Implementacija - FB Forensics Viewer**
 - Opis, namena i primena
 - Skup podataka, arhitektura aplikacije, ključne funkcionalnosti
 - Integracija gotovih AI modela
- **Zaključak**

Domeni

- **Forenzika društvenih medija**
- Kompjuterska forenzika
- Mrežna forenzika
- Forenzika mobilnih uređaja
- Forenzika baza podataka
- Cloud forenzika



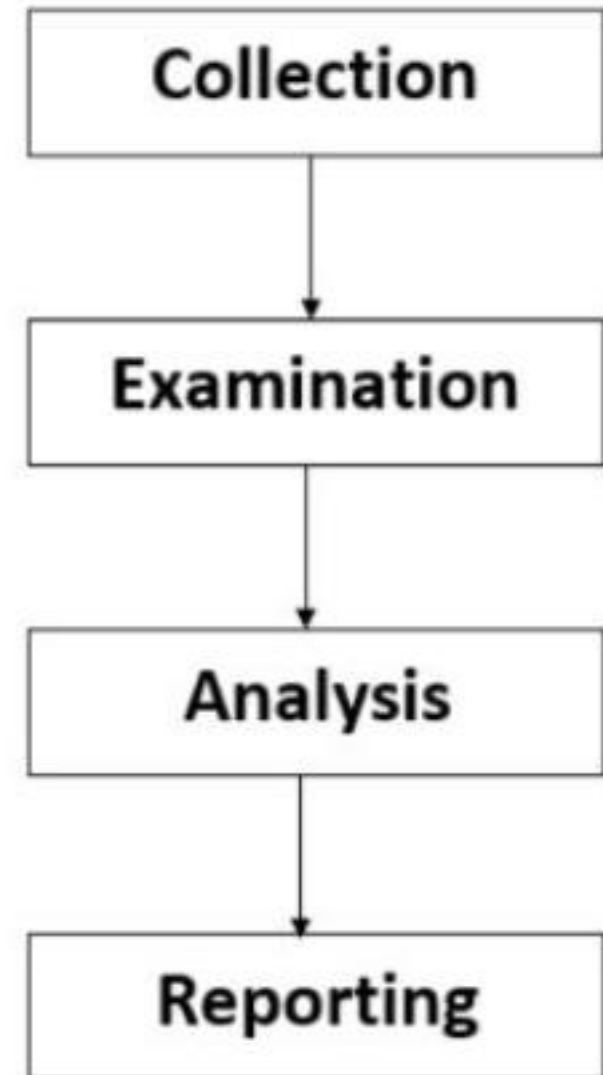
Digitalna forenzika društvenih medija

- Proces vađenja i analize digitalnih dokaza sa platformi društvenih medija kako bi se podržale pravne istrage
- Ispitivanje različitih elemenata, kao što su postovi, poruke, liste prijatelja, korisnički profili i metapodaci, kako bi se otkrile vredne informacije koje mogu pomoći u rešavanju zločina, proveriti alibija ili utvrđivanju kredibiliteta osobe
- **Ciljevi:**
 - Rekonstruisanjem vremenskih linija, identifikovanje ključnih aktera i razumevanje konteksta određenih situacija
 - Identifikacija kriminalnih aktivnosti
 - Otkrivanje skrivenih mreža
 - Analiza javnog mnjenja i raspoloženja
 - Alati: Detekcija raspoloženja, dipfejkova, forenzika slika i videa, detekcija anomalija



Faze forenzike društvenih medija

- Prikupljanje
- Ispitivanje
- Analiza
- Izveštavanje

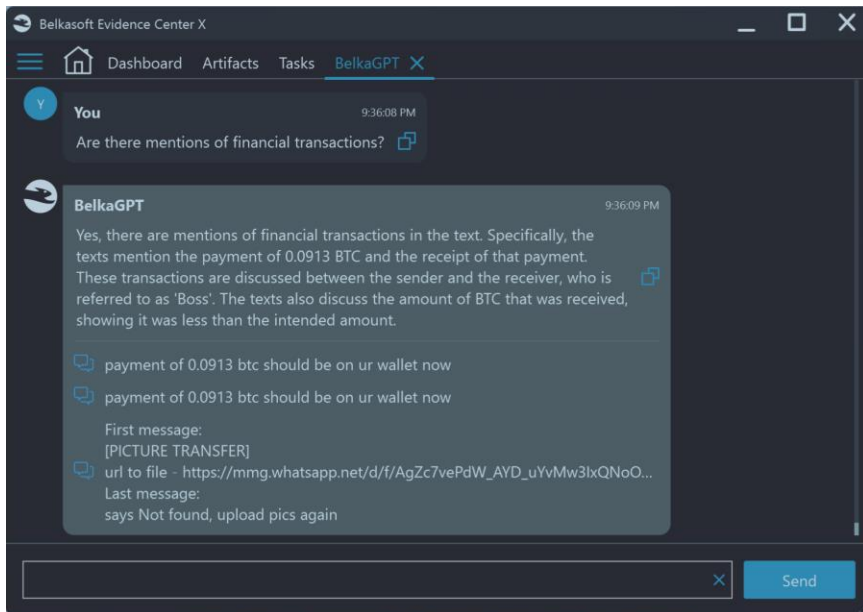


Alati za istraživanje društvenih medija

- **Jahteon** - pomaže u arhiviranju društvenih medija, usklađenosti sa društvenim medijima i otkrivanju podataka
- **WebPreserver** - prikupljanje i očuvanje društvenih medija i druge internet imovine
- **Pipl** - prikuplja podatke iz javnih dokumenata, malih oglasa, listinga, imenika i onlajn arhiva na internetu, ali ima i svoje ekskluzivne izvore podataka
- **Social Links**
- **TinEye** - praćenje originalnih slika i obrnutih pretraga slika
- **Oxygen Forensic Detective** - prikupljanje i analizu podataka sa mobilnih uređaja, cloud servisa i online naloga
- **Belkasoft X** - obrada komunikacije, multimedijalnih fajlova, istorije aktivnosti i drugih tragova relevantnih za istragu

OXYGEN FORENSIC
Detective

Uticaj veštačke inteligencije na digitalnu forenziku društvenih medija



- Veštačka inteligencija ne samo da unapređuje digitalnu forenziku, već istovremeno redefiniše njene osnovne principe
- Automatizacija analize velikih količina podataka
- AI uvodi i nizove novih izazova: Pojava sintetičkog sadržaja, uključujući takozvanu deepfake tehnologiju i generisani tekst, koji može imitirati autentičnu komunikaciju korisnika
- NLP u forenzici društvenih medija
- GAN-ovi u forenzici društvenih medija
- Izazovi: Ograničen pristup podacima uz očuvanje privatnosti korisnika, nedovoljna objašnjivost modela, rizik od pristrasnosti i potreba za stalnim prilagođavanjem usled brzog razvoja tehnologije i promena u ponašanju korisnika
- BelkaSoft i BelkaGPT



Facebook digitalna forenzika

- Vremenska linija, prijatelji, najvažnije vesti, Pages, igre, grupe, događaji, Messenger, Stream
- Pravila
- Poseban izazov zbog heterogenosti podataka, njihove raspoređenosti između više izvora i čestih promena u strukturi aplikacija i servisa.
- Pored tekstualne komunikacije, predmet analize mogu biti i multimedijalni sadržaji, lokacijski podaci, kontakti, sesije prijavljivanja, interakcije sa objavama i drugi oblici korisničkog ponašanja.
- U praksi retko oslanja na jedan alat, već se koristi kombinacija više softverskih rešenja.

Implementacija – FB Forensics Viewer

- Analiza podataka izdvojenih iz arhive Facebook korisničkog naloga
- Pregled, organizacija, analiza i interpretaciju digitalnih artefakata na način koji je prilagođen potrebama digitalno-forenzičke obrade.
- Ova aplikacija može poslužiti:
 - Krajnjim korisnicima da provere bezbednost svog naloga
 - Digitalnim forenzičarima u preliminarnoj analizi kompromitovanih naloga
 - Edukativne svrhe – studenti informatike, sajber bezbednosti i digitalne forenzike
 - Sudski veštaci i istražitelji, kao pomoćno sredstvo u analiziranju online aktivnosti

Archive successfully loaded

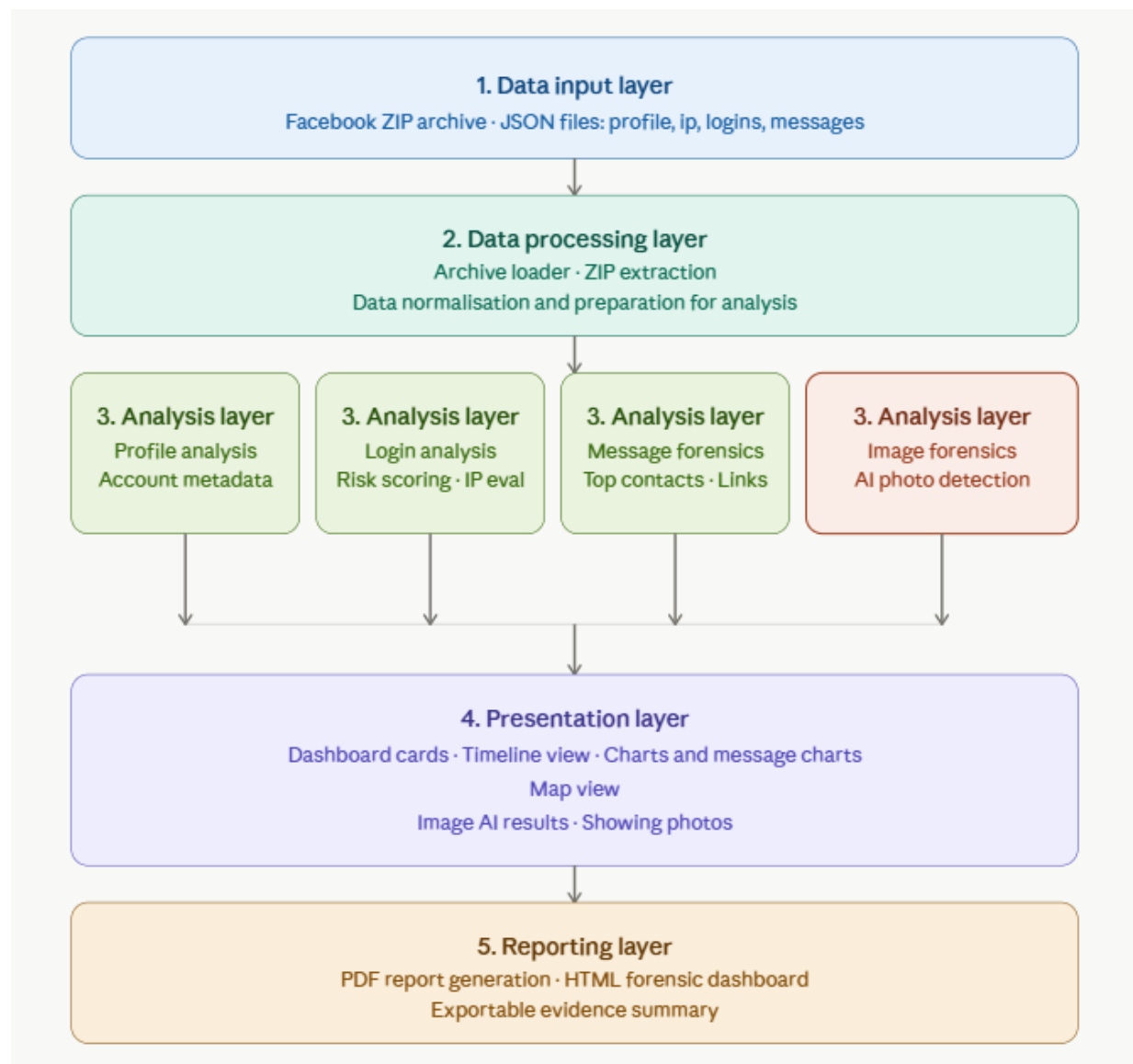
Email	Account created	Archive
marina.testovic@example.com	2017-04-18	Load

Location: ALL Search:

Date	Location	IP Address	
	Niš, Serbia	109.245.18.201	HIGH
	Vienna, Austria	62.99.14.8	MEDIUM
	Vienna, Austria	62.99.14.8	LOW
	Niš, Serbia	109.245.18.201	LOW
	Belgrade, Serbia	178.149.22.10	MEDIUM
	Budapest, Hungary	81.183.20.14	MEDIUM
	Budapest, Hungary	81.183.20.14	LOW
	Vienna, Austria	62.99.14.8	LOW
	Vienna, Austria	62.99.14.8	LOW
	Novi Sad, Serbia	188.2.45.77	MEDIUM
	Belgrade, Serbia	178.149.22.10	LOW
	Prague, Czech Republic	185.75.44.90	MEDIUM
	Budapest, Hungary	81.183.20.14	LOW
	Belgrade, Serbia	178.149.22.10	LOW
	Novi Sad, Serbia	188.2.45.77	LOW

Arhitektura, skup podataka, biblioteke

- Desktop aplikacija - **Python**
- Korišćenje standardnih Python biblioteka:
 - za obradu podataka (zipfile, json, os),
 - biblioteka za GUI (tkinter), vizualizaciju (matplotlib),
 - kao i naprednih AI alata (scikit-learn, PyTorch, Hugging Face Transformers) za analizu i detekciju anomalija.
 - Generisanje izveštaja realizovano je pomoću biblioteke reportlab.



Prikaz rezultata

account_activity.jsonmessage_1.json

DatotekaUrediPrikazati

```
{
  "participants": [
    "Marina Testovic",
    "Marko Nikolic"
  ],
  "messages": [
    {
      "sender_name": "Marko Nikolic",
      "timestamp_ms": 1709429040000,
      "content": "Hitno, javi se kad vidiš poruku."
    },
    {
      "sender_name": "Marina Testovic",
      "timestamp_ms": 1709429460000,
      "content": "Videla sam, šta se desilo?"
    },
    {
      "sender_name": "Marko Nikolic",
      "timestamp_ms": 1709429580000,
      "content": "Ništa strašno, samo prover mail."
    },
    {
      "sender_name": "Marko Nikolic",
      "timestamp_ms": 1717807440000,
      "content": "Pošalji kod, pošalji kod, pošalji kod."
    }
  ]
}
```

account_activity.jsonmessage_1.json

DatotekaUrediPrikazati

```
{
  "profile": {
    "name": "Marina Marinkovic",
    "email": "marina.marinkovic@example.com",
    "account_created": "2017-04-18"
  },
  "logins": [
    {
      "date": "2024-01-05",
      "location": "Niš, Serbia",
      "ip": "109.245.18.201"
    },
    {
      "date": "2024-01-10",
      "location": "Vienna, Austria",
      "ip": "62.99.14.8"
    },
    {
      "date": "2024-01-15",
      "location": "Vienna, Austria",
      "ip": "62.99.14.8"
    },
    {
      "date": "2024-01-20",
      "location": "Niš, Serbia",
      "ip": "109.245.18.201"
    }
  ]
}
```

FB - Forensics Viewer

FB - Forensics Viewer

Digital forensic analysis of Facebook archive

Archive successfully loaded (fb_archive_root_only_large.zip)

Name

Marina Testovic

Email

marina.testovic@example.com

Account created

2017-04-18

Archive status

Loaded

Modules

Logins

AI Analysis

Timeline

Charts

Messages

Message Forensics

Message charts

Map

Login analysis

Risk: ALLLocation: ALLSearch:

Reset filters

Date	Location	IP Address	Risk
2024-01-05	Niš, Serbia	109.245.18.201	HIGH
2024-01-10	Vienna, Austria	62.99.14.8	MEDIUM
2024-01-15	Vienna, Austria	62.99.14.8	LOW
2024-01-20	Niš, Serbia	109.245.18.201	LOW
2024-01-25	Belgrade, Serbia	178.149.22.10	MEDIUM
2024-01-30	Budapest, Hungary	81.183.20.14	MEDIUM
2024-02-04	Budapest, Hungary	81.183.20.14	LOW
2024-02-09	Vienna, Austria	62.99.14.8	LOW
2024-02-14	Vienna, Austria	62.99.14.8	LOW
2024-02-19	Novi Sad, Serbia	188.2.45.77	MEDIUM
2024-02-24	Belgrade, Serbia	178.149.22.10	LOW
2024-02-29	Prague, Czech Republic	185.75.44.90	MEDIUM
2024-03-05	Budapest, Hungary	81.183.20.14	LOW
2024-03-10	Belgrade, Serbia	178.149.22.10	LOW
2024-03-15	Novi Sad, Serbia	188.2.45.77	LOW

FB - Forensics Viewer

Digital forensic analysis of Facebook archive

Load Archive

Export Report

Archive successfully loaded (fb_archive_root_only_large.zip)

Name

Marina Testovic

Email

marina.testovic@example.com

Account created

2017-04-18

Archive status

Loaded

Modules

Logins

AI Analysis

Timeline

Charts

Messages

Message Forensics

Message charts

Map

AI analysis

Reason: New location, New IP address

=== RISK SCORE ===

Total account score: 610

Risk level of the profile: HIGH

[HIGH] (70) 2024-01-05 | Niš, Serbia | IP:109.245.18.201
Reasons: First record (+10), New location (+40), New IP (+20)

[MEDIUM] (60) 2024-01-10 | Vienna, Austria | IP:62.99.14.8
Reasons: New location (+40), New IP (+20)

[LOW] (0) 2024-01-15 | Vienna, Austria | IP:62.99.14.8
Reasons:

[LOW] (0) 2024-01-20 | Niš, Serbia | IP:109.245.18.201
Reasons:

[MEDIUM] (60) 2024-01-25 | Belgrade, Serbia | IP:178.149.22.10
Reasons: New location (+40), New IP (+20)

[MEDIUM] (60) 2024-01-30 | Budapest, Hungary | IP:81.183.20.14
Reasons: New location (+40), New IP (+20)

[LOW] (0) 2024-02-04 | Budapest, Hungary | IP:81.183.20.14
Reasons:

[LOW] (0) 2024-02-09 | Vienna, Austria | IP:62.99.14.8

Loggins modul

Modules

Logins

AI Analysis

Image Forensics

Timeline

Charts

Messages

Message Forensics

Message charts

Map

Login analysis

Risk:

LOW

Location:

ALL

Search:

Serbia

Reset filters

Date	Location	IP Address	Risk
2024-01-20	Niš, Serbia	109.245.18.201	LOW
2024-02-24	Belgrade, Serbia	178.149.22.10	LOW
2024-03-10	Belgrade, Serbia	178.149.22.10	LOW
2024-03-15	Novi Sad, Serbia	188.2.45.77	LOW
2024-03-20	Novi Sad, Serbia	188.2.45.77	LOW
2024-03-25	Belgrade, Serbia	178.149.22.10	LOW
2024-03-30	Novi Sad, Serbia	188.2.45.77	LOW
2024-04-29	Belgrade, Serbia	178.149.22.10	LOW
2024-06-03	Niš, Serbia	109.245.18.201	LOW

Procena rizika naloga

FB - Forensics Viewer

Digital forensic analysis of Facebook archive

Load Archive

Export Report

Archive successfully loaded (fb_archive_root_only_large.zip)

Name

Marina Testovic

Email

marina.testovic@example.com

Account created

2017-04-18

Archive status

Loaded

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Reasons: New location (+40), New IP (+20)

[LOW] (0) 2024-01-15 | Vienna, Austria | IP:62.99.14.8

Reasons:

[LOW] (0) 2024-01-20 | Niš, Serbia | IP:109.245.18.201

Reasons:

[MEDIUM] (60) 2024-01-25 | Belgrade, Serbia | IP:178.149.22.10

Reasons: New location (+40), New IP (+20)

[MEDIUM] (60) 2024-01-30 | Budapest, Hungary | IP:81.183.20.14

Reasons: New location (+40), New IP (+20)

[LOW] (0) 2024-02-04 | Budapest, Hungary | IP:81.183.20.14

Reasons:

[LOW] (0) 2024-02-09 | Vienna, Austria | IP:62.99.14.8

Reasons:

Modul za analizu slika - AI modeli

FB - Forensics Viewer

Digital forensic analysis of Facebook archive

Load Archive

Export Report

Archive successfully loaded (fb_archive_with_image_samples.zip)

Name

Marina Testovic

Email

marina.testovic@example.com

Account created

2017-04-18

Archive status

Loaded

Modules

Logins

AI Analysis

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Message charts

Map

Image forensics

Image forensic analysis

Preliminary AI-based check for potentially manipulated or AI-generated images.

Scan archive images

Analyze found images

Prediction: ALL

Source: ALL

Search:

File name	Source	Prediction	Confidence (%)	Status
suspicious_render_01.png	Media	Potentially AI-generated	99.99	Review suggested
travel_real_01.jpg	Media	Potentially AI-generated	100.0	Review suggested
message_attachment_01.jpg	Messages	Potentially AI-generated	99.23	Review suggested
message_attachment_ai_style_01.png	Messages	Potentially AI-generated	99.77	Review suggested
profile_ai_style_01.png	Profile	Potentially AI-generated	100.0	Review suggested
profile_real_01.jpg	Profile	Potentially AI-generated	99.97	Review suggested

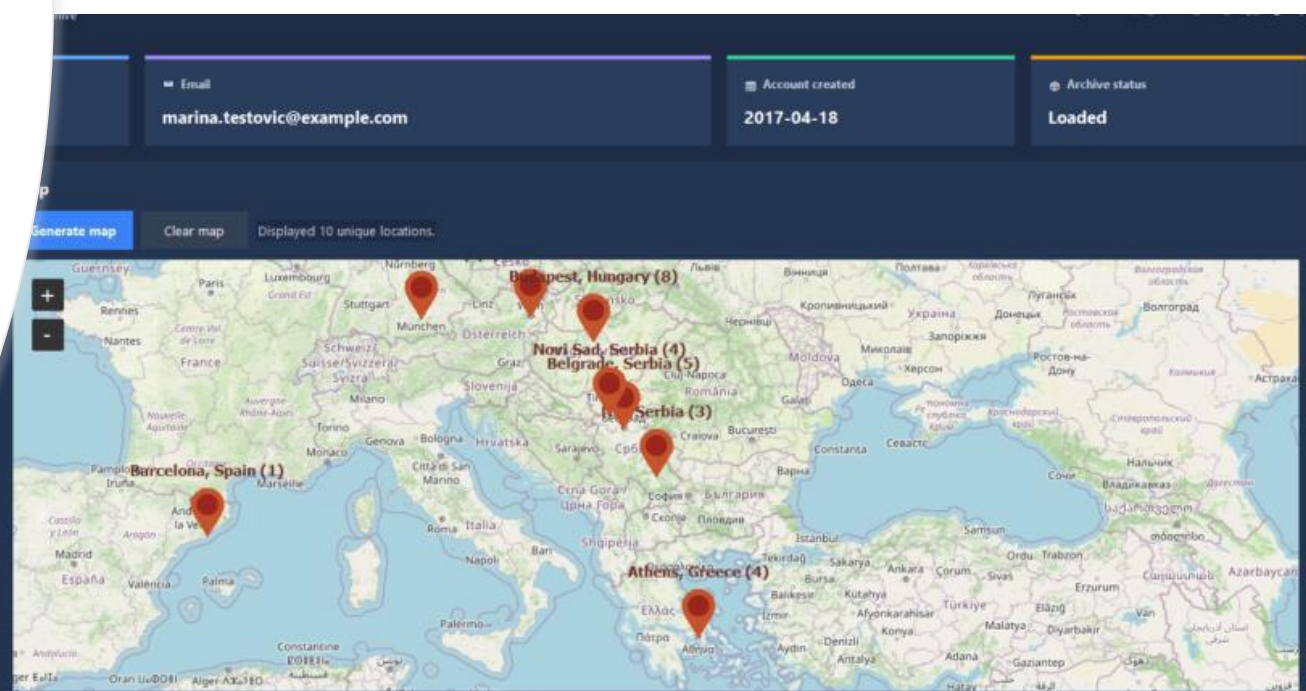
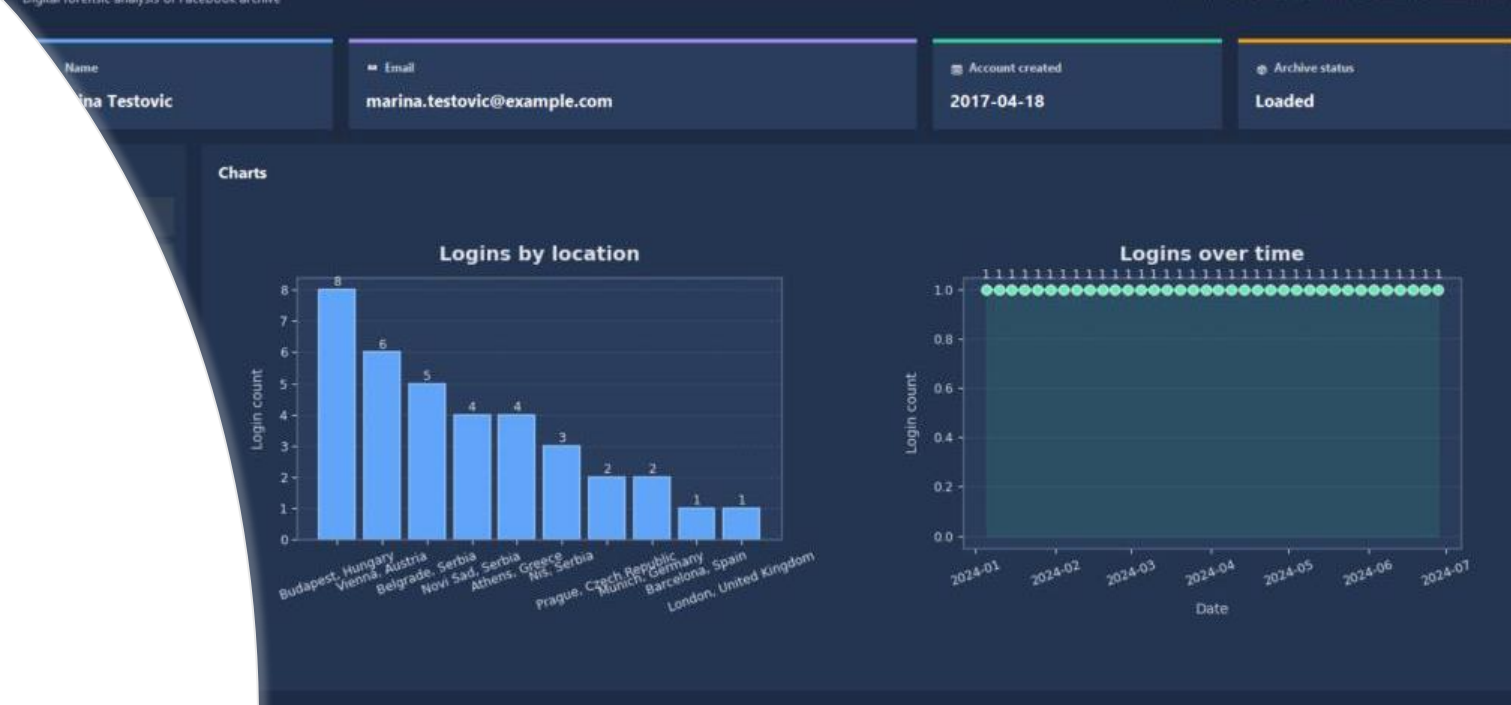
Image preview

Shared invoice screenshot

PAID

Open original

Geografski prikaz prijavljivanja, prikaz rezultata uz pomoć grafikona



Generisanje izveštaja

Facebook Forensics Report

Case: Facebook Account Analysis | Examiner: FB - Forensics Viewer | Report Generated: 2026-04-14 22:58:51

Source file: Loaded archive | Account: Marina Testovic | Email: marina.testovic@example.com | Account created: 2017-04-18

Summary of Findings

684

Messages Parsed

5

Key Contacts

10

Unique Locations

1

Suspicious Indicators

HIGH

Account Risk Level

Flagged Messages

Sender	AI Categories	Message Snippet
Marina Testovic	Urgency / Pressure	Šaljem odmah.
Marko Nikolic	Urgency / Pressure	Hitno, javi se kad vidiš poruku.
Marko Nikolic	Credential Request	Pošalji kod, pošalji kod, pošalji kod.
Marko Nikolic	Credential Request	Pošalji kod, pošalji kod, pošalji kod.
Marko Nikolic	Credential Request	Pošalji kod, pošalji kod, pošalji kod.

Risky Activities

Date	Activity	Risk Level
2024-01-05	Login from Niš, Serbia IP: 109.245.18.201	HIGH
2024-01-10	Login from Vienna, Austria IP: 62.99.14.8	HIGH
2024-01-15	Login from Vienna, Austria IP: 62.99.14.8	HIGH
2024-01-20	Login from Niš, Serbia IP: 109.245.18.201	HIGH
2024-01-25	Login from Belgrade, Serbia IP: 178.149.22.10	HIGH
2024-01-30	Login from Budapest, Hungary IP: 81.183.20.14	HIGH
2024-02-04	Login from Budapest, Hungary IP: 81.183.20.14	HIGH
2024-02-09	Login from Vienna, Austria IP: 62.99.14.8	HIGH

Evidence Collected

- Profile information extracted
- Login history analyzed
- Message threads parsed
- Timeline generated
- Charts generated
- AI-assisted message analysis completed

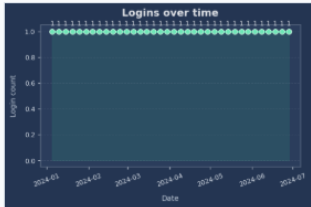
AI Summary

The conversation analysis covered 15 text messages. Detected semantic categories include: Credential Request, Urgency / Pressure. 8 messages were sent during night hours. Repeated message patterns were identified, which may indicate spam, pressure tactics, or automation. Overall AI-assisted conversation risk was assessed as HIGH.

Detected categories:

- Credential Request
- Urgency / Pressure

Charts



Timeline Preview

- No timeline lines available.

Key Contacts

- M

Marina Testovic

3 messages
- A

Ana Petrovic

1 messages
- M

Marko Nikolic

1 messages
- I

Ivana

1 messages
- M

Milan

1 messages

Suspicious Links / Bursts

- Links
 - No suspicious links detected.
- Burst Events
 - No burst activity detected.

Note: Account risk level is HIGH, message risk level is LOW. Additional analyst review is recommended for medium and high risk indicators.

Ideje za dalji razvoj i istraživački pravci

- Proširenje podrške za kompletnu Facebook arhivu
- Automatsko parsiranje + verzionisanje strukture podataka
- Forenzička pouzdanost: **chain-of-custody, hash, audit log**
- Napredni timeline i geolokacijska analiza
- AI/ML: detekcija anomalija, analiza ponašanja i sadržaja
- Semantička pretraga, sažimanje i izdvajanje ključnih informacija
- Modularna arhitektura + plugin sistem
- Integracija sa drugim platformama i razvoj AI asistenta



Zaključak

- Forenzika društvenih mreža predstavlja ključan alat u savremenim istragama, jer omogućava efikasno prikupljanje i analizu digitalnih dokaza, uz značajno ubrzanje procesa i smanjenje troškova.

- Primena veštačke inteligencije dodatno unapređuje analizu velikih količina podataka, ali zahteva odgovornu upotrebu zasnovanu na etičkim principima i objašnjivosti rezultata.

- Razvijena aplikacija **FB - Forensics Viewer** potvrđuje da je moguće integrisati forenzičke metode, vizuelizaciju i AI pristupe u jedinstven sistem, koji predstavlja dobru osnovu za dalji razvoj profesionalnih alata.

Account created

2017-04-18

Archive status

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Daily Search:

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HVALA NA PAŽNJI!

Marijana Cvetković, br. Indeksa 1431